AI Interpretation and Bias

Matthew Engelhard

Investigating model predictions

Looking inside the black box

When expert and model disagree…

You've been using a new AI model that predicts 2-year risk of progression to stage IV chronic kidney disease. Up to now, you've found its predictions to be helpful as well as consistent with your clinical judgement. This model is based on electronic health record data, including diagnosis and procedure codes, medications, labs, and vitals; as well as social determinants of health.

You see a 32 year-old male patient with new significant hypertension (165/91). You check your model and see it predicts low stage IV CKD risk (<2%). However, on physical exam there is puffiness around his eyes and 1+ edema to his mid-shin, and on urinalysis you find significant proteinuria (3+). These findings concern you, and you believe the patient's true risk is much higher than the model is saying.

<u>Report Back</u>: What can you do to investigate the discrepancy between the model's prediction and your clinical judgement?

Key Questions

- What information is available to the model?
  - Predictor groups
  - Timing of collection
  - Lab *ID* or lab *values*?

- What population was used to train the model?
  - Relevant inclusion / exclusion criteria
  - Representation of specific demographics

- <u>What do you know that the model doesn't (or vice versa)?</u>

Report Back:
What else should you know to use the model?

Model Facts **Model name:** Deep Sepsis **Locale:** Duke University Hospital

Approval Date: 09/22/2019 **Last Update:** 01/13/2020 **Version:** 1.0

Summary

This model uses EHR input data collected from a patient's current inpatient encounter to estimate the probability that the patient will meet sepsis criteria within the next 4 hours. It was developed in 2016-2019 by the Duke Institute for Health Innovation. The model was licensed to Cohere Med in July 2019.

Mechanism

- **Outcome** ……sepsis within the next 4 hours, see outcome definition in "Other Information"
- **Output** ……0% - 100% probability of sepsis occurring in the next 4 hours
- **Target population** ……all adult patients >18 y.o. presenting to DUH ED
- **Time of prediction** ……every hour of a patient's encounter
- **Input data source**……electronic health record (EHR)
- **Input data type** ……demographics, analytes, vitals, medication administrations
- **Training data location and time-period** ……DUH, diagnostic cohort, 10/2014 – 12/2015
- **Model type**…… Recurrent Neural Network

Validation and performance

	Prevalence	AUC	PPV @ Sensitivity of 60%	Sensitivity @ PPV of 20%	Cohort Type	Cohort URL / DOI
Local Retrospective	18.9%	0.88	0.14	0.50	Diagnostic	arxiv.org/abs/1708.05894
Local Temporal	6.4%	0.94	0.20	0.66	Diagnostic	jmir.org/preprint/15182
Local Prospective	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
External	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Target Population	6.4%	0.94	0.20	0.66	Diagnostic	jmir.org/preprint/15182

Uses and directions

- **Benefits:** Early identification and prompt treatment of sepsis can improve patient morbidity and mortality.
- **Target population and use case:** Every hour, data is pulled from the EHR to calculate risk of sepsis for every patient at the DUH ED. A rapid response team nurse reviews every high-risk patient with a physician in the ED to confirm whether or not to initiate treatment for sepsis.
- **General use:** This model is intended to be used to by clinicians to identify patients for further assessment for sepsis. The model is not a diagnostic for sepsis and is not meant to guide or drive clinical care. This model is intended to complement other pieces of patient information related to sepsis as well as a physical evaluation to determine the need for sepsis treatment.
- **Appropriate decision support:** The model identifies patient X as at a high risk of sepsis. A rapid response team nurse discusses the patient with the ED physician caring for the patient and they agree the patient does not require treatment for sepsis.
- **Before using this model:** Test the model retrospectively and prospectively on a diagnostic cohort that reflects the target population that the model will be used upon to confirm validity of the model within a local setting.
- **Safety and efficacy evaluation:** Analysis of data from clinical trial (NCT03655626) is underway. Preliminary data shows rapid response team, nurse-driven workflow was effective at improving sepsis treatment bundle compliance.

Warnings

- **Risks:** Even if used appropriately, clinicians using this model can misdiagnose sepsis. Delays in a sepsis diagnosis can lead to morbidity and mortality. Patients who are incorrectly treated for sepsis can be exposed to risks associated with unnecessary antibiotics and intravenous fluids.
- **Inappropriate Settings:** This model was not trained or evaluated on patients receiving care in the ICU. Do not use this model in the ICU setting without further evaluation. This model was trained to identify the first episode of sepsis during an inpatient encounter. Do not use this model after an initial sepsis episode without further evaluation.
- **Clinical Rationale:** The model is not interpretable and does not provide rationale for high risk scores. Clinical end users are expected to place model output in context with other clinical information to make final determination of diagnosis.
- **Inappropriate decision support:** This model may not be accurate outside of the target population, primarily adults in the non-ICU setting. This model is not a diagnostic and is not designed to guide clinical diagnosis and treatment for sepsis.
- **Generalizability:** This model was primarily evaluated within the local setting of Duke University Hospital. Do not use this model in an external setting without further evaluation.
- **Discontinue use if:** Clinical staff raise concerns about utility of the model for the indicated use case or large, systematic changes occur at the data level that necessitates re-training of the model.

Other information:

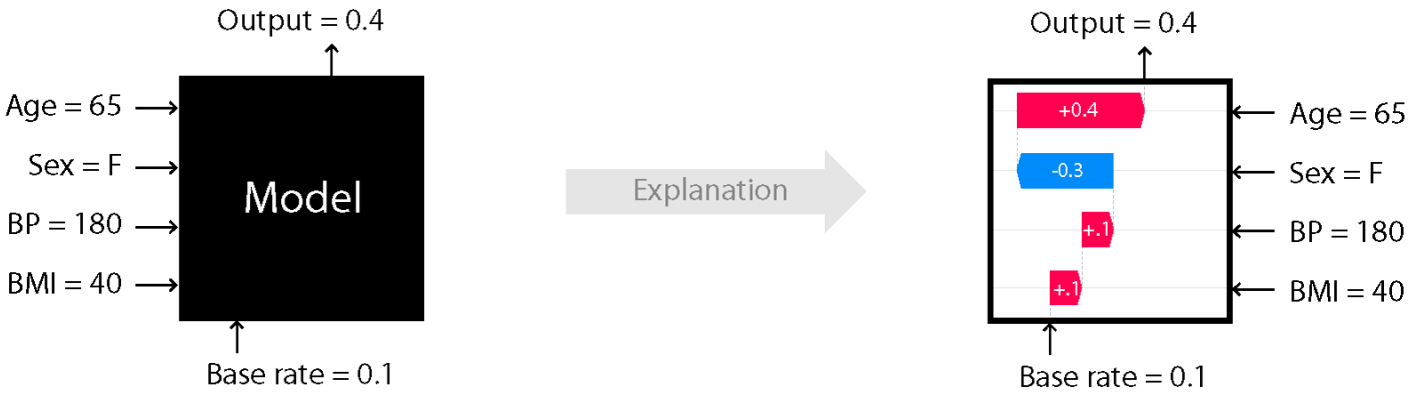
- **Outcome Definition:** https://doi.org/10.1101/648907
- **Related model:** http://doi.org/10.1001/jama.2016.0288
- **Model development & validation:** arxiv.org/abs/1708.05894
- **Model implementation**: jmir.org/preprint/15182
- **Clinical trial:** clinicaltrials.gov/ct2/show/NCT03655626
- **Clinical impact evaluation:** TBD
- **For inquiries and additional information:** please email mark.sendak@duke.edu

Sendak, M.P., Gao, M., Brajer, N. *et al.* Presenting machine learning model information to clinical end users with model facts labels. *npj Digit. Med.* **3**, 41 (2020). https://doi.org/10.1038/s41746-020-0253-3

Understanding Model Predictions

- What variables are available to the model, and when were they collected?
- What variables are influencing the model's prediction?
- What findings are influencing your impression?

SHAP



Algorithmic Bias

What is it, where does it come from, and what can be done?

Growing concerns about algorithmic bias

You've been using the model for a year now, and you've noticed something potentially concerning.

The model seems to preferentially refer White patients over Black patients. Specifically, you notice that Black patients are referred only if their clinical presentation is more severe compared to White patients who are referred. You are not sure why this is happening.

<u>Report Back</u>: What are some possible sources of this bias, and what could you do to investigate?

Sources of Algorithmic Bias

Many possible sources, but in most cases, it's:

1. Biased outcomes in training data
2. Poor representation of demographics

JOURNAL ARTICLE

Observability and its impact on differential bias for clinical prediction models Get access

Mengying Yan, Michael J Pencina, L Ebony Boulware, Benjamin A Goldstein

Journal of the American Medical Informatics Association, Volume 29, Issue 5, May 2022, Pages 937–943, https://doi.org/10.1093/jamia/ocac019

Published: 25 February 2022 **Article history** ▾

What does it mean for a model to be *fair*?

Consider whether predictions are fair between two groups A and B.

One possible solution is *fairness through unawareness*, also called *blindness,* wherein the model is not given information about group status.

<u>Report back</u>:

- What could go wrong if the model is *aware* of group status?
- What could go wrong if the model is *unaware* of group status?

Then, propose an alternative approach to achieve or evaluate fairness.

Science

FEATURE

A troubled calculus

Researchers use race to build disease risk assessment tools. Can removing it help resolve medicine's race crisis?

JYOTI MADHUSOODANAN Authors Info & Affiliations

SCIENCE • 23 Jul 2021 • Vol 373, Issue 6553 • pp. 380-383 • DOI: 10.1126/science.373.6553.380

636 4

PERSPECTIVE

Health Economics WILEY

Patient-centered appraisal of race-free clinical risk assessment

Charles F. Manski

Department of Economics, Institute for Policy Research, Northwestern University, Evanston, Illinois, USA

Correspondence
Charles F. Manski, Department of Economics, Institute for Policy Research, Northwestern University, Evanston, IL, USA.
Email: cfmanski@northwestern.edu

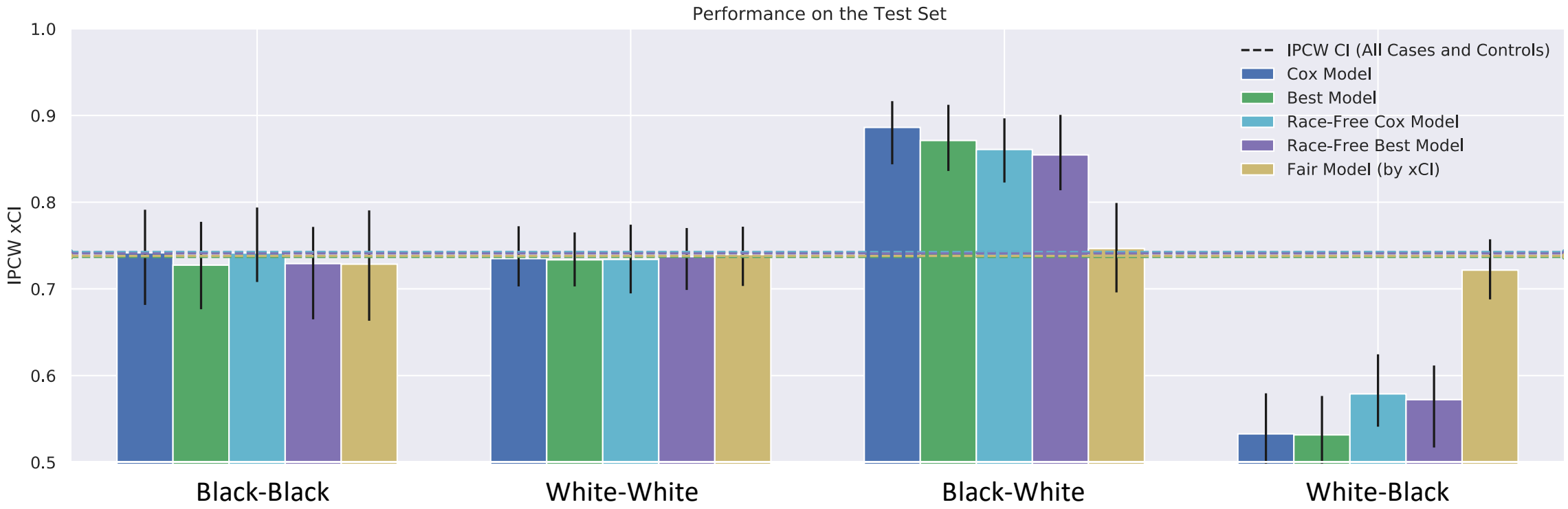
Abstract

Until recently, there has been a consensus that clinicians seeking to assess patient risks of illness should condition risk assessments on all observed patient covariates with predictive power. The broad idea is that knowing more about patients enables more accurate predictions of their health risks and, hence, better clinical decisions. This consensus has recently unraveled with respect to a specific covariate, namely race. There have been increasing calls for race-free risk assessment, arguing that using race to predict health risks contributes to racial disparities and inequities in health care. In some medical fields, leading institutions have recommended race-free risk assessment. An important open question is how race-free risk assessment would affect the quality of clinical decisions. Considering the matter from the patient-centered perspective of medical economics yields a disturbing conclusion: Race-free risk assessment would harm patients of all races.

KEYWORDS
predicting illness risk, racial health disparities, risk assessment algorithms

Blindness can be ineffective or even harmful.

- In some cases it is effective.
- In other cases it isn't enough to fix the problem.
- In still other cases, it can make the problem worse.



Engelhard et al. Exploring trade-offs in equitable stroke risk prediction with parity-constrained and race-free models. Artificial Intelligence in Medicine, 2025.

Possible definitions of algorithmic fairness

Consider whether predictions are fair between two groups A and B.

1. Blindness: The model *cannot see* group status
2. Independence: Model predictions are *independent* of group status
3. Separation: Model *performance* is similar in all groups
4. Individual fairness: The model makes *similar predictions* for similar individuals regardless of group status

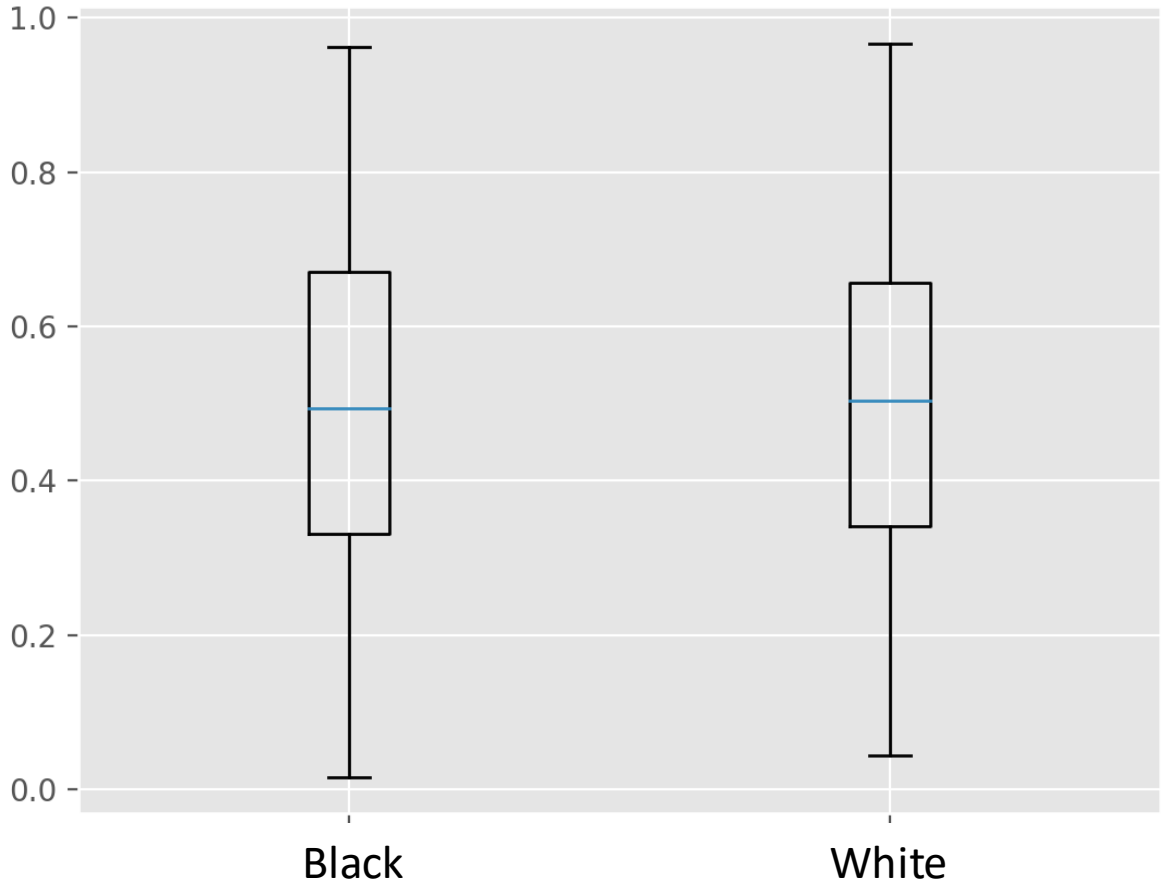
Except under special (& not realistic) circumstances, (1) – (4) are mutually exclusive.

Zooming in on *Independence*

Model predictions are independent of group status.

This implies that the distribution of model predictions is the same in each group.

Is this **fair**?

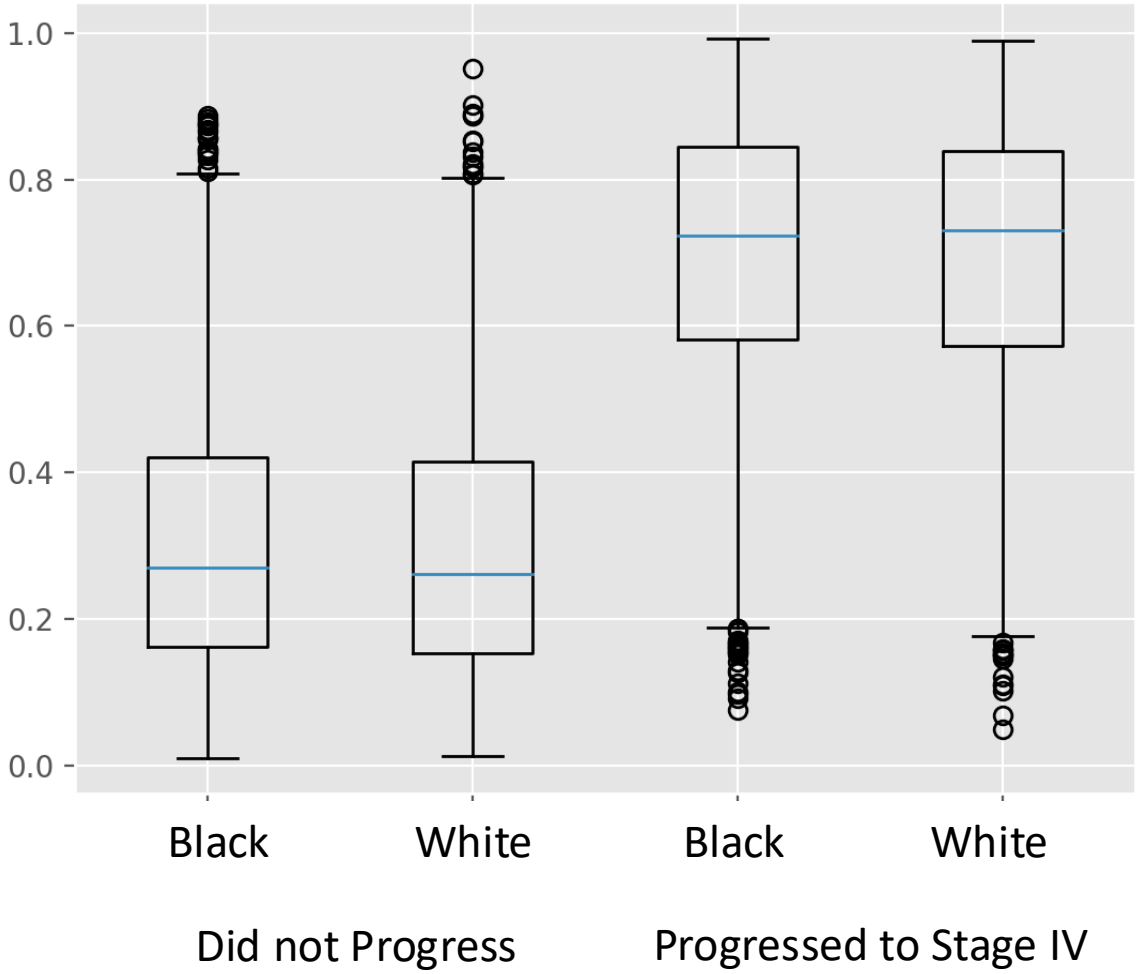


Zooming in on *Separation*

Model predictions are independent of group status *after conditioning on the outcome*.

This implies when considering individuals *with the same outcome*, the distribution of model predictions is the same in each group.

Is this **fair**?



Possible definitions of algorithmic fairness

Consider whether predictions are fair between two groups A and B.

1. Blindness: The model *cannot see* group status
2. Independence: Model predictions are *independent* of group status
3. Separation: Model *performance* is similar in all groups
4. Individual fairness: The model makes *similar predictions* for similar individuals regardless of group status

Except under special (& not realistic) circumstances, (1) – (4) are mutually exclusive.

Algorithmic Bias: A Wicked Problem?

1. No definitive formulation
2. Problem is ongoing
3. Difficult to define or test success
4. Possible solutions have effects that may not be reversable
5. Can be considered to be a symptom of another problem
6. Many explanations for the problem
7. No right or wrong answers

Rittell and Webber, 1973

How do we
- evaluate
- understand
- debias
generative AI?

Will genAI help address biases, or exacerbate them?

ME

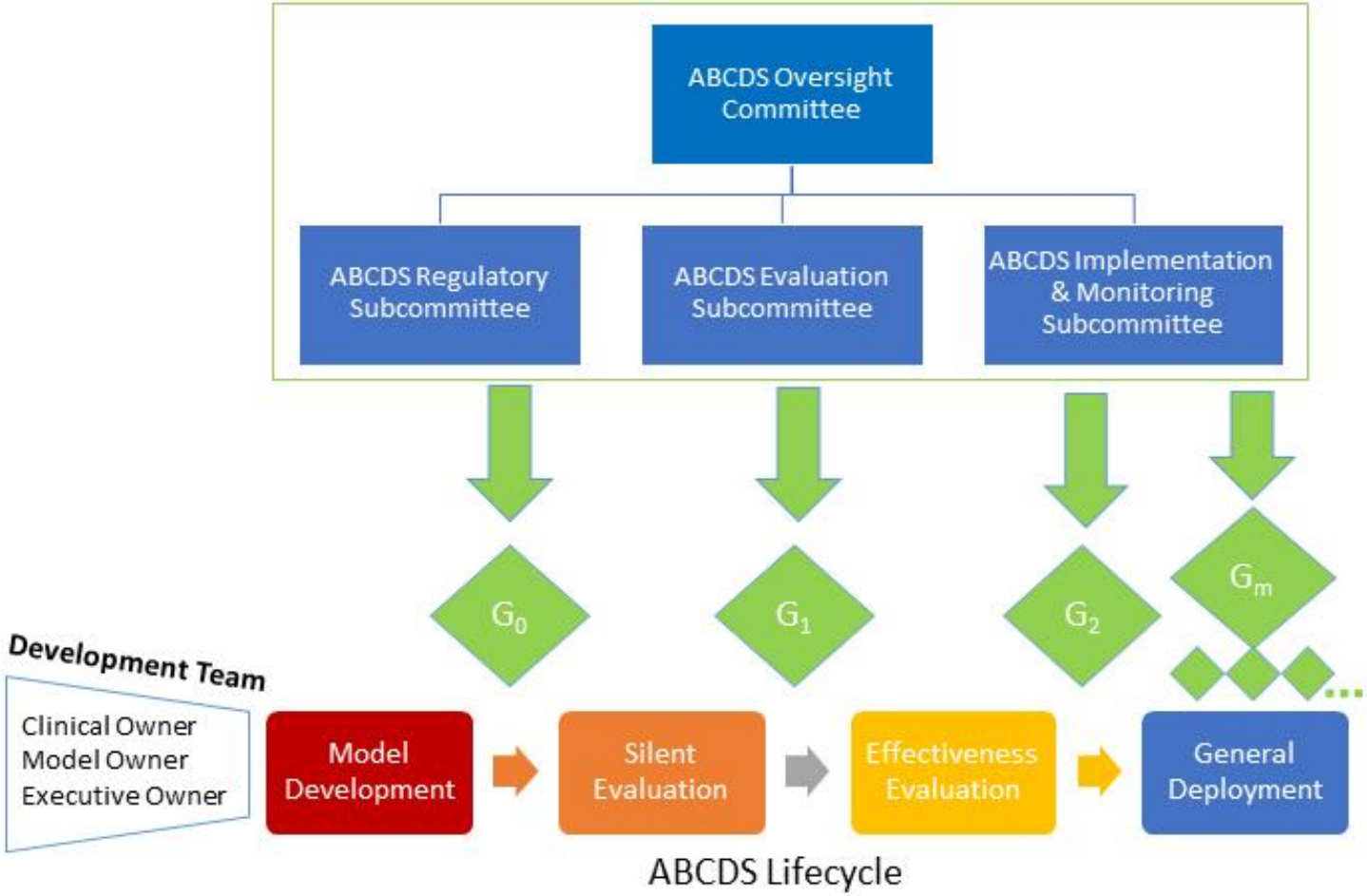
I'm giving a lecture to our first-year medical students introducing them to AI and its applications to clinical medicine. What should I tell them about natural language processing, and about you (ChatGPT) specifically?

Natural Language Processing (NLP) is a subfield of artificial intelligence that focuses on enabling machines to understand and process human language. It has numerous applications in clinical medicine, including but not limited to clinical documentation, medical coding, clinical decision support, and patient communication.

As for ChatGPT, you can introduce me as a state-of-the-art NLP model trained by OpenAI. I can answer a wide range of questions and engage in natural language conversations with humans, which has numerous potential applications in the medical field. For example, I can be used to provide virtual assistants for patients to answer their medical questions, to help physicians with documentation and coding, and to analyze large amounts of medical records to identify patterns and insights that might not be immediately apparent to human analysts.

However, it's important to note that while AI and NLP models like ChatGPT have great potential, they are not a substitute for human expertise in the medical field. It's important to use these technologies in conjunction with human clinicians to improve patient care and outcomes.

Regenerate response

ChatGPT Feb 13 Version. Free Research Preview. Our goal is to make AI systems more natural and safe to interact with. Your feedback will help us improve.

Toward *Trustworthy* CDS Development